

Exercise 4 Analysis

Early Stage Analysis (Before 10 000 evaluations)

In nearly every progress plot, the Genetic Algorithm (GA) curve starts with a sharp initial rise within the first thousand evaluations, indicating rapid exploitation of promising regions. Refer to appendix 1, the GA's steep evaluation during the first 5000 evaluations shows that crossover and mutation effectively combine high-quality individuals to accelerate development. The Evolutionary Algorithm (EA) and the Randomised Local Search (RLS) have similar starts to the GA but plateau earlier, as shown in appendix 2, where their curve flattens at around the 10 000 evaluation mark. In contrast, the MOEA and GSEMO initially start with a slower trajectory, impacted by population diversity and multi objective selection pressure that prioritises broad exploration rather than immediate optimisation.

Mid and Late Stage Analysis (At and After 10 000 evaluations)

Appendix 3 shows the divergence among algorithms after the 10 000 evaluation point. GA achieves the largest average improvement, at around +799 on average, closely followed by MOEA at +736. These consistent gains suggest that both these algorithms maintain diversity deep into the run, preventing premature convergence. The EA and RLS algorithms have more moderate progressions at +400 and +380 respectively, showing smaller but refined ones once their early growth starts to decline. SOEA and GSEMO have the smallest growths at +10 to +30, their stagnation being evident when their progress curves are analysed. Across appendix 4, GA again shows consistent late stage growth, while MOEA climbs at a slower pace initially, but maintaining its continuous nature, showing strong long term behaviour.

Differences Between MaxCoverage and MaxInfluence Problems

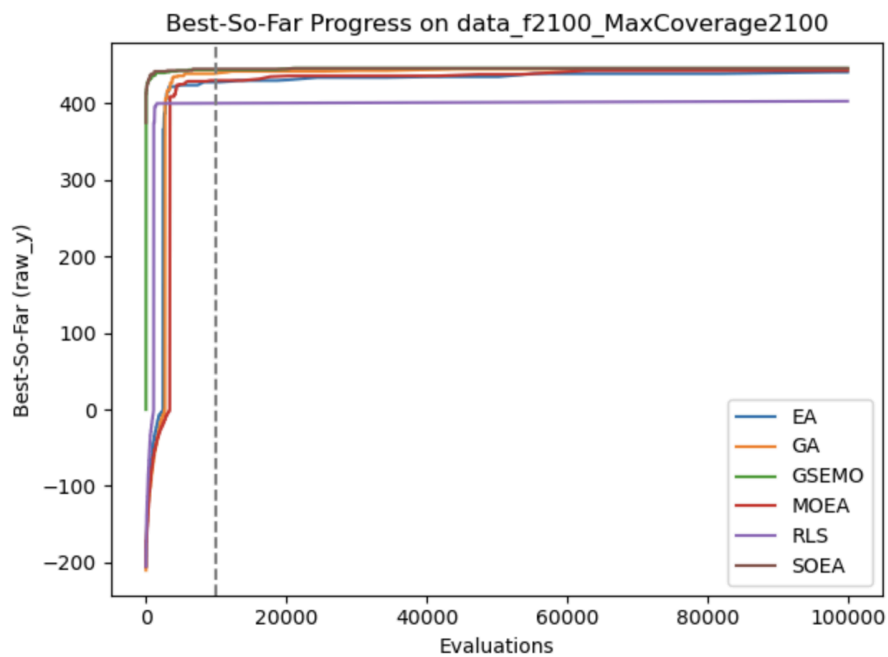
While the performance hierarchy remains consistent across problem types, the two problems highlight different strengths between the algorithms. Appendices 1 and 2 show the GA's advantage of recombining partial coverage solutions, while MOEA benefits from balancing sub-objectives like redundancy and coverage speed. In Appendix 4, influence propagation introduces more stochasticity, giving EA and RLS occasional dominance when GA and MOEA temporarily converge. However, even in these scenarios, GA's steady and greedy improvements after 10 000 evaluations highlight it as the most reliable algorithm.

Overall Interpretation

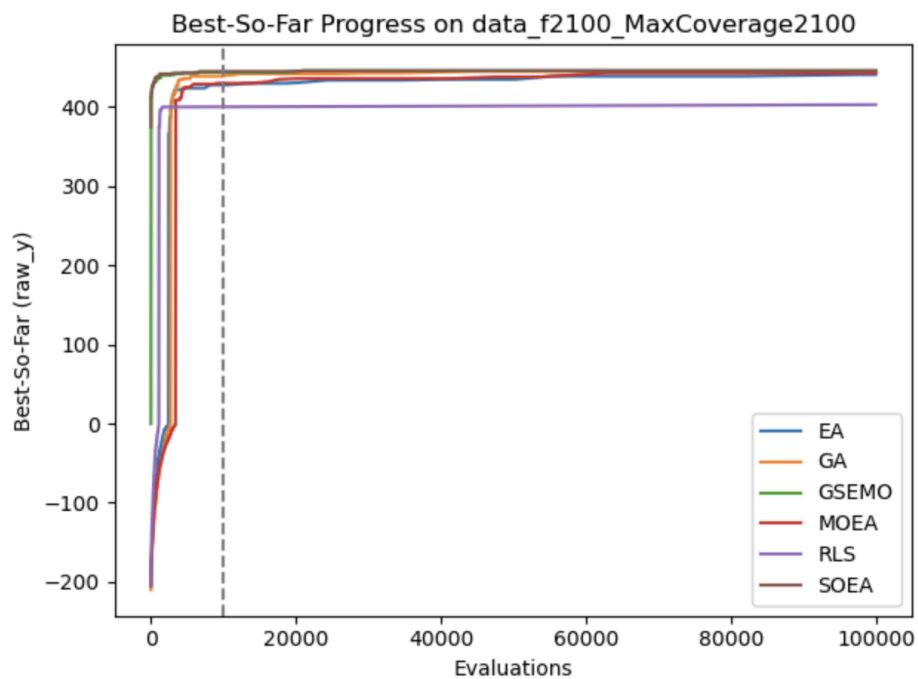
- GA exhibits both rapid early development and continued long term progress, yielding the best final values across nearly all plots.
- MOEA remains competitive, trailing slightly but maintains consistent momentum across the performance range.
- EA and RLS perform well under smaller budgets, but their decline after 10 000 evaluations show limited exploratory power.
- GSEMO and SEOA converge early and exhibit little improvement afterwards.

Appendices

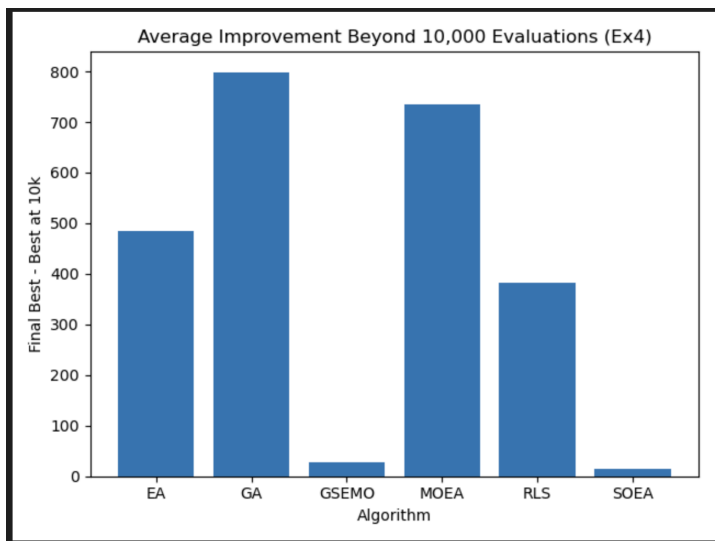
Appendix 1 - Max Coverage 2100 Graph



Appendix 2 - Max Coverage 2101 Graph



Appendix 3 - Algorithm Improvement Graph



Appendix 4 - Max Influence 2202 Graph

